

Maximal Podleś Lipschitz derivatives need not lie in $SU_q(2)$

A counterexample and a complete diagonal characterization

Automated mathematical discovery run `fa_banach_001`

13 August 2026

Status. Candidate counterexample and complete negative answer, likely valid, subject to human review. The result also characterizes the range on the whole commutative algebra $C^*(A)$.

1 Source question

Aguilar and KAAD study the Dąbrowski–Sitarz spectral triple on the standard Podleś sphere in [1]. On the coordinate algebra, the commutator has off-diagonal entries ∂_1 and ∂_2 taking values in $SU_q(2)$. They extend these maps to the maximal Lipschitz algebra as bounded operators $H_+ \rightarrow H_-$ and $H_- \rightarrow H_+$, respectively, and ask:

“We do not know whether the images of ∂_1 and ∂_2 are contained in the C^* -algebra $SU_q(2)$ even though this holds for their restrictions to the coordinate algebra $\mathcal{O}(S_q^2)$.”

This is Section 3, official PDF page 7. The complete source passage is shown in Figure 1.

Theorem 3.1 (Dąbrowski–Sitarz). *The triple $(\mathcal{O}(S_q^2), H_+ \oplus H_-, D_q)$ is an even spectral triple.*

We let $\text{Lip}_{D_q}(S_q^2) \subseteq S_q^2$ denote the Lipschitz algebra associated to the Dirac operator $D_q : \text{Dom}(D_q) \rightarrow H_+ \oplus H_-$ and record the formula

$$d_q(x) = \begin{pmatrix} 0 & \partial_2(x) \\ \partial_1(x) & 0 \end{pmatrix} \quad x \in \mathcal{O}(S_q^2)$$

for the associated derivation. In particular we obtain extensions of ∂_1 and ∂_2 to the Lipschitz algebra $\text{Lip}_{D_q}(S_q^2)$. These two extensions are again denoted by ∂_1 and ∂_2 :

$$\partial_1 : \text{Lip}_{D_q}(S_q^2) \rightarrow \mathbb{L}(H_+, H_-) \quad \partial_2 : \text{Lip}_{D_q}(S_q^2) \rightarrow \mathbb{L}(H_-, H_+).$$

We do not know whether the images of ∂_1 and ∂_2 are contained in the C^* -algebra $SU_q(2)$ even though this holds for their restrictions to the coordinate algebra $\mathcal{O}(S_q^2)$.

4. MATRIX UNITS FOR THE STANDARD PODLEŚ SPHERE

Figure 1: The two maximal extensions and the range question in [1, p. 7].

2 Definitions and statement

Fix $q \in (0, 1)$. Recall that $SU_q(2)$ is generated by a, b with

$$ba = qab, \quad b^*a = qab^*, \quad bb^* = b^*b, \quad a^*a + q^2bb^* = 1 = aa^* + bb^*.$$

The standard sphere $S_q^2 \subseteq SU_q(2)$ is generated by $A = b^*b$ and $B = ab^*$. Put

$$p_k = \chi_{\{q^{2k}\}}(A), \quad k \in \mathbb{N}_0.$$

Thus $C^*(A) \cong C(\{0\} \cup \{q^{2k} : k \geq 0\})$. For $f \in C(\text{Sp}(A))$, write

$$f_k = f(q^{2k}), \quad f_\infty = f(0), \quad w_k(f) = \frac{q^{-k}\sqrt{1-q^{2k+2}}}{1-q^2}(f_k - f_{k+1}). \quad (2.1)$$

We use the source's identification of an element of $SU_q(2)$ with its left multiplication operator between the homogeneous GNS subspaces.

Theorem 1 (Diagonal domain and range characterization). *For every $f \in C(\text{Sp}(A))$:*

1. $f(A) \in \text{Lip}_{D_q}(S_q^2)$ if and only if $w(f) = (w_k(f))_{k \geq 0} \in \ell^\infty$.
2. When these conditions hold,

$$\partial_1(f(A)) \in SU_q(2) \iff w(f) \in c_0.$$

Consequently the source question has a negative answer for both ∂_1 and ∂_2 .

An explicit common witness for both maps is the selfadjoint element $x = f(A)$ defined by $f(0) = 0$ and

$$f(q^{2k}) = \sum_{j=k}^{\infty} \frac{(1-q^2)q^j(-1)^j}{\sqrt{1-q^{2j+2}}}. \quad (2.2)$$

Indeed, $w_k(f) = (-1)^k$.

3 Proof intuition

On $C^*(A)$, the derivation is an ordinary finite-difference operator in disguise. The factor in (2.1) is exactly the norm normalization that turns its mutually orthogonal columns into a weighted backward shift. Bounded weights therefore describe the *maximal* commutator domain. Membership in the original C^* -algebra is stricter.

In the faithful Toeplitz representation of $SU_q(2)$, the derivative becomes

$$\sum_{k \geq 0} w_k E_{k,k+1} \otimes U^*.$$

A Toeplitz weighted shift must have convergent weights. The quotient of $SU_q(2)$ is independent of the second circle variable, while the displayed operator retains U^* ; hence the limit must actually be zero. This is the gap between bounded commutators and the graph closure of the coordinate derivation.

4 Proof

We use two computations from [1]. First,

$$\partial_1(p_k) = \frac{q^{-2k}p_k - q^{-2(k-1)}p_{k-1}}{1 - q^2} b^* a^*, \quad (4.1)$$

with the second term absent for $k = 0$ (Lemma 5.3 there). Second, their Proposition 6.7 gives the right-column formula for every element of the maximal Lipschitz algebra. Specializing it to $f(A)$ gives

$$(1 - q^2)\partial_1(f(A))p_{k+1} = (f_k - f_{k+1})q^{-3k-2}\sqrt{1 - q^{2k+2}} \cdot f_{k,k+1}(b^*)^2, \quad (4.2)$$

where $f_{r,s}$ are the source's matrix units. Proposition 6.2 there says

$$\|f_{k,k+1}(b^*)^2\| = q^{2k+2}.$$

It follows immediately from (4.2) that

$$\|\partial_1(f(A))p_{k+1}\| = |w_k(f)|. \quad (4.3)$$

The maximal domain

If $f(A)$ is Lipschitz, (4.3) proves that $w(f)$ is bounded. Conversely, suppose $\|w(f)\|_\infty = M < \infty$. Telescoping (2.1) yields

$$f_k - f_\infty = (1 - q^2) \sum_{j=k}^{\infty} \frac{q^j w_j(f)}{\sqrt{1 - q^{2j+2}}}. \quad (4.4)$$

In particular, $f_k - f_\infty = O(q^k)$.

Define the finite truncations

$$x_N = f_\infty 1 + \sum_{k=0}^N (f_k - f_\infty) p_k.$$

Then $x_N \rightarrow f(A)$ in norm. All interior normalized derivative weights of x_N are $w_0, \dots, w_{N-1}$. Its sole boundary weight is

$$\beta_N = \frac{q^{-N} \sqrt{1 - q^{2N+2}}}{1 - q^2} (f_N - f_\infty).$$

By (4.4), monotonicity of $1 - q^{2j+2}$ gives

$$|\beta_N| \leq M \sum_{j=N}^{\infty} q^{j-N} = \frac{M}{1 - q}. \quad (4.5)$$

The same bound holds for the other off-diagonal derivative because $\partial_2(z) = -\partial_1(z^*)^*$. Hence the commutator seminorms of x_N are uniformly bounded. The commutator seminorm is lower semicontinuous [1, Lemma 2.3], so $f(A) \in \text{Lip}_{D_q}(S_q^2)$. This proves part 1.

The C^* -range

We next record the concrete obstruction. Let U be multiplication by the coordinate function on $L^2(\mathbb{T})$, let (e_k) be the standard basis of $\ell^2(\mathbb{N}_0)$, and consider the standard faithful representation ρ of $SU_q(2)$ [2, Chapter 4]:

$$\begin{aligned}\rho(a)(e_k \otimes \xi) &= \sqrt{1 - q^{2k+2}} e_{k+1} \otimes \xi, \\ \rho(b)(e_k \otimes \xi) &= q^k e_k \otimes U\xi.\end{aligned}\tag{4.6}$$

Thus $\rho(p_k) = E_{k,k} \otimes 1$, and a direct use of the source's normalization of $f_{k,k+1}$ gives

$$\rho\left(\frac{(f_k - f_{k+1})q^{-3k-2}\sqrt{1 - q^{2k+2}}}{1 - q^2} f_{k,k+1}(b^*)^2\right) = w_k(f) E_{k,k+1} \otimes U^*.\tag{4.7}$$

Suppose $\partial_1(f(A))$ were left multiplication by some $y \in SU_q(2)$. Equation (4.2) then determines yp_{k+1} as the element inside (4.7); the $k = 0$ case of the source's full column formula also gives $\partial_1(f(A))p_0 = 0$. There is no faithfulness gap here. If $z \in SU_q(2)$ acts as zero on H_+ , then in particular $za = zb = 0$, because $a, b \in \mathcal{A}_1 \subset H_+$; faithfulness of the Haar representation and $aa^* + bb^* = 1$ give

$$z = z(aa^* + bb^*) = (za)a^* + (zb)b^* = 0.\tag{4.8}$$

Since $\sum p_k = 1$ strongly in (4.6), all columns together force

$$\rho(y) = W_w \otimes U^*, \quad W_w = \sum_{k=0}^{\infty} w_k E_{k,k+1}.\tag{4.9}$$

We now identify exactly when (4.9) belongs to $\rho(SU_q(2))$. Let $\mathcal{T} = C^*(S)$ be the Toeplitz algebra. Since the weights of $\rho(a)$ tend to one, $\rho(a) - S \otimes 1$ is compact in the first factor. Spectral projections of $\rho(A)$, powers of $\rho(b)$, and powers of $\rho(a)$ generate every $E_{ij} \otimes U^m$. Therefore

$$\rho(SU_q(2)) = C^*(S \otimes 1, \mathcal{K} \otimes C(\mathbb{T})) = \{X \in \mathcal{T} \otimes C(\mathbb{T}) : (\sigma \otimes \text{id})(X) \in C(\mathbb{T}) \otimes 1\},\tag{4.10}$$

where $\sigma : \mathcal{T} \rightarrow C(\mathbb{T})$ is the Toeplitz symbol map. This also proves the needed pullback description directly from (4.6).

If (4.9) belongs to (4.10), evaluation in the second circle variable shows $W_w \in \mathcal{T}$. Every $T \in \mathcal{T}$ has the form $T_g + K$ with $g \in C(\mathbb{T})$ and K compact. Hence $\langle T e_{k+1}, e_k \rangle$ converges to the corresponding Fourier coefficient of g . Applied to W_w , this proves $w_k \rightarrow L$ for some $L \in \mathbb{C}$. Then $W_w - LS^*$ is compact and

$$(\sigma \otimes \text{id})(W_w \otimes U^*) = L \bar{\zeta} \otimes \bar{u}.\tag{4.11}$$

The pullback condition in (4.10) says that this function is independent of u , so $L = 0$. Thus $w \in c_0$ is necessary.

If $w \in c_0$, then W_w is compact, so $W_w \otimes U^* \in \mathcal{K} \otimes C(\mathbb{T}) \subseteq \rho(SU_q(2))$. Equivalently, the orthogonal-column series in (4.2) converges in norm, and its limit is the maximal derivative. Therefore $w \in c_0$ is also sufficient. Part 2 is proved.

The explicit witness and both derivatives

The series in (2.2) converges absolutely and is $O(q^k)$, so it defines a real continuous f on $\text{Sp}(A)$ with $f(0) = 0$. Its successive difference is its first summand, and (2.1) gives $w_k(f) = (-1)^k$. Theorem 1 now gives

$$x = f(A) \in \text{Lip}_{D_q}(S_q^2), \quad \partial_1(x) \notin SU_q(2).$$

Since $x = x^*$ and $\partial_2(x) = -\partial_1(x)^*$, membership of $\partial_2(x)$ in the $*$ -algebra $SU_q(2)$ would imply membership of $\partial_1(x)$. Hence $\partial_2(x) \notin SU_q(2)$ as well. This completes the negative answer. $\square$

5 Verification and limitations

The accompanying script checks (2.1)–(2.2), the finite weighted-shift matrix entries, and the uniform boundary bound (4.5) for four deformation parameters. These checks audit indexing and normalization only; the proof is the analytic argument above.

The result concerns the maximal Lipschitz algebra used in the source. It is compatible with the fact that the graph closure of the coordinate derivation does take values in $SU_q(2)$: on the diagonal slice that smaller graph domain is exactly the additional condition $w \in c_0$.

A bounded literature search on 13 August 2026 used local run indexes and fresh searches for the exact quoted sentence, the title and authors, “Podleś Lipschitz algebra”, “maximal Lipschitz”, “derivative range”, and “ $SU_q(2)$ ”. It found no later explicit answer to the inclusion question. The directly relevant later paper [4] distinguishes the graph-closure algebra from the maximal Lipschitz algebra and explains that maximal derivatives generally require a von Neumann-algebraic framework. Its Theorem B approximates in C^* -norm with a uniform Lip bound, not in graph norm; it neither states the original $SU_q(2)$ question nor gives the explicit counterexample or the ℓ^∞/c_0 characterization above. It therefore strongly anticipates the negative mechanism. Novelty confidence is moderate-low for the qualitative yes/no answer and moderate for the precise diagonal theorem.

The separate question in the source introduction about $q \rightarrow 1$ convergence was answered in [3]; that does not overlap the result here.

The highest-priority human checks are (4.7), including its phase, and the faithfulness step (4.8). The Toeplitz pullback argument after those points is standard and elementary.

References

- [1] K. Aguilar and J. Kaad, *The Podleś sphere as a spectral metric space*, J. Geom. Phys. 133 (2018), 260–278, doi:10.1016/j.geomphys.2018.07.015, [arXiv:1803.03027](#).
- [2] A. Klimyk and K. Schmüdgen, *Quantum Groups and Their Representations*, Texts and Monographs in Physics, Springer, 1997.
- [3] K. Aguilar, J. Kaad, and D. Kyed, *The Podleś spheres converge to the sphere*, Comm. Math. Phys. 392 (2022), 1029–1061, [arXiv:2102.12761](#).
- [4] K. Aguilar, J. Kaad, and D. Kyed, *Polynomial approximation of quantum Lipschitz functions*, Documenta Math. 27 (2022), 765–787, [arXiv:2104.04317](#).